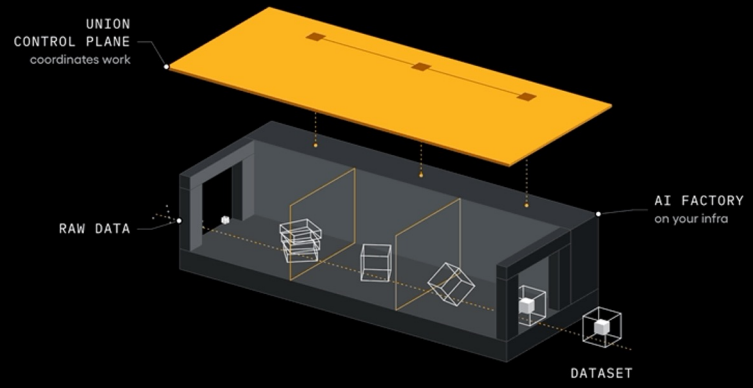
Prompt, fine-tune, and RL

Customizing NVIDIA Nemotron with Union on AWS

Michael Galarnyk, Developer Relations, Union.ai · November 10, 2026



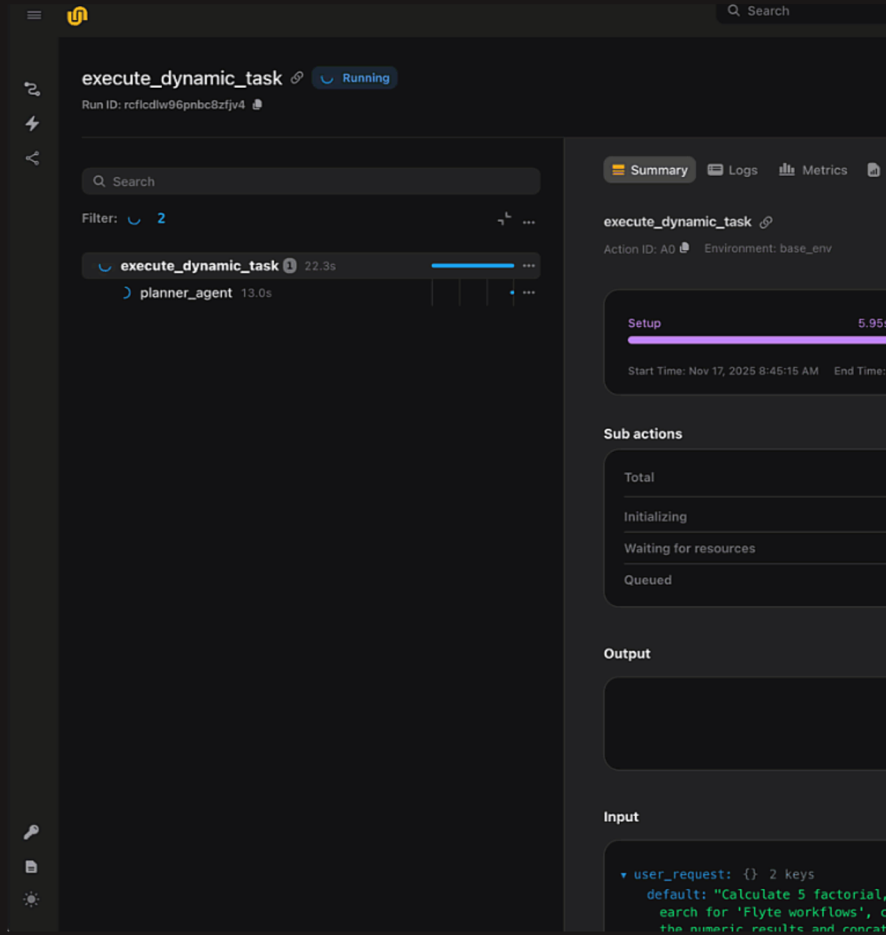
Union in one minute

What: The runtime that orchestrates AI pipelines, models, and agents.

How: Workloads heal themselves when the infrastructure fails.

Who: Toyota, Prima Mente, Artera, Lockheed Martin, Mistral. 4,000+ companies on Union or open-source Flyte.

Where: In the customer's own cloud. Data never leaves it.



Use case: will the US Fed raise interest rates?

One 8-minute Fed speech wiped ~$3 trillion off US stocks in a day.

1 Label every sentence

hawkish toward raising rates	"Market-based measures of inflation compensation have also picked up."
dovish toward cutting rates	"I think that keeping interest rates low is still appropriate for our economy."
neutral no lean	"Our current framework for implementing monetary policy is working very well."

→

2 Score each Fed release

(hawkish – dovish) ÷ all sentences

3 Use the score

Tracks inflation and Treasury yields. Trading on it beat buy-and-hold, 2011–2022.

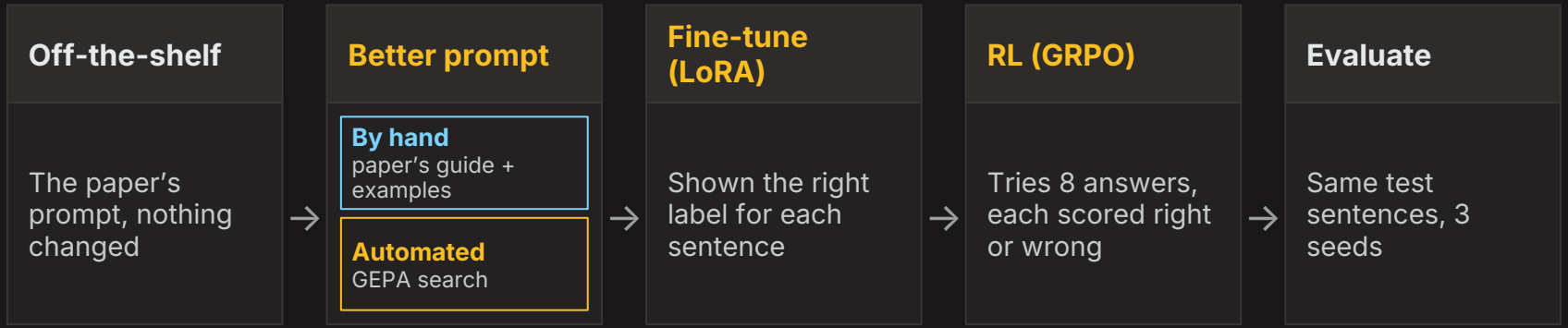
Speech: Powell, Jackson Hole, Aug 26, 2022 · Sentences from Fed minutes, press conferences, speeches · Shah et al. 2023

Ways to customize Nemotron 3 Nano 4B

	MANUAL by a person	**AUTOMATED** learned from data		
	Manual prompt engineering	**Prompt optimization (GEPA)**	**Fine-tuning (LoRA)**	**Reinforcement learning (GRPO)**
What changes	**The prompt**	**The prompt**	**The weights**	**The weights**
Learns from	A person's domain knowledge	Its own mistakes	The right answers, shown to it	A score on its own tries

Each can build on the one before

Our workflow: one Flyte pipeline



Each step is a Flyte task: a failed step retries and resumes where it stopped.

Off-the-shelf: the paper’s prompt

STARTING POINT · Shah et al. 2023’s prompt, verbatim

System prompt

Discard all the previous instructions. Behave like you are an expert sentence classifier. Classify the following sentence from FOMC into **'HAWKISH'**, **'DOVISH'**, or **'NEUTRAL'** class. Label **'HAWKISH'** if it is corresponding to tightening of the monetary policy, **'DOVISH'** if it is corresponding to easing of the monetary policy, or **'NEUTRAL'** if the stance is neutral. Provide the label in the first line and provide a short explanation in the second line.

Manual prompt engineering with domain knowledge

```
Classify the following sentence from FOMC into 'HAWKISH', 'DOVISH', or 'NEUTRAL'
class.
```

← task description

```
Labeling guide (Shah et al. 2023, Table 10):
Economic Status. Dovish: when inflation decreases, when unemployment increases, …
Hawkish: when inflation increases, when unemployment decreases, …
Labor. Dovish: when productivity increases. Hawkish: when productivity decreases.
…
```

← **domain knowledge**

```
Sentence: The persistence of underutilized resources was expected to foster some
moderation in core price inflation.
Label: NEUTRAL
Sentence: So unemployment has tended to go up much faster for minorities and for
others who are—tend to be at the low end of the income spectrum.
Label: DOVISH
```

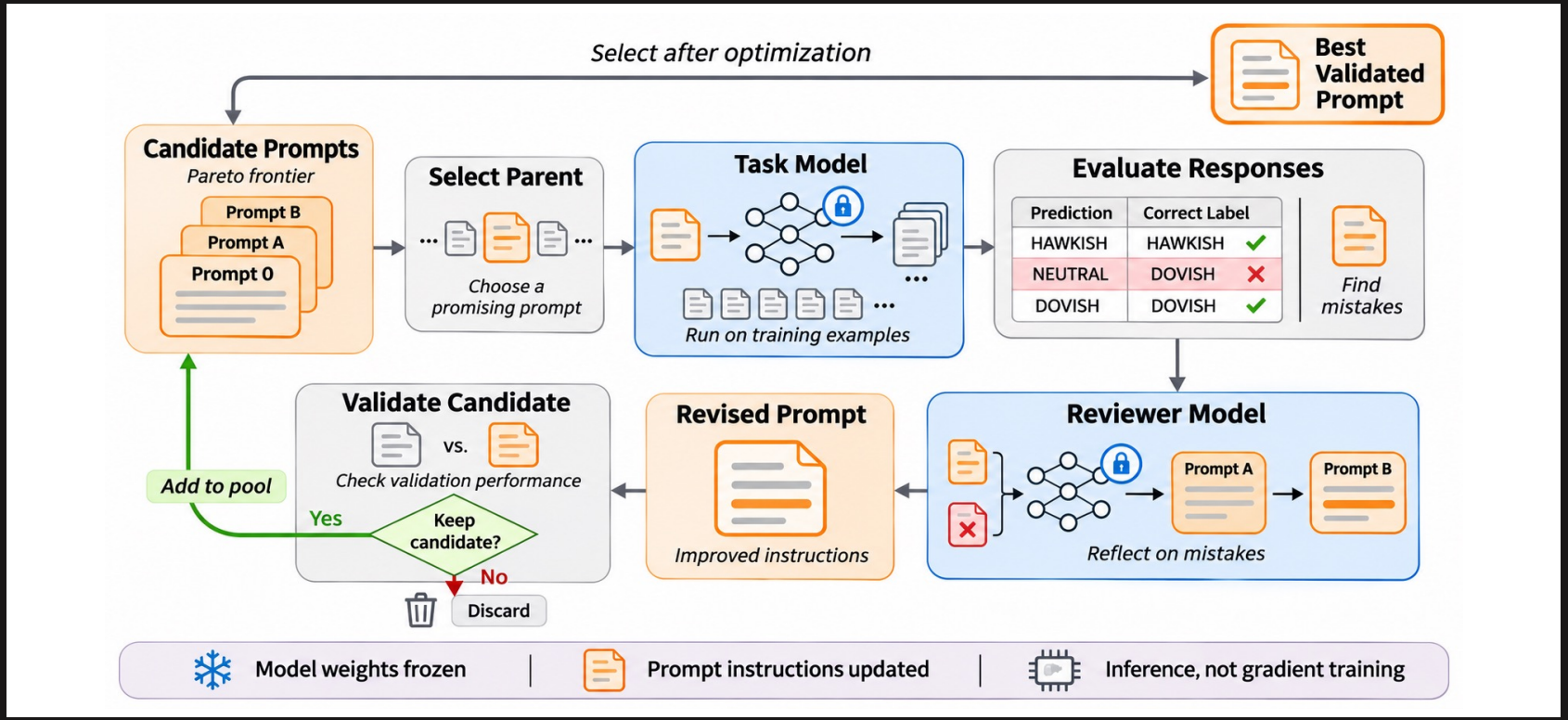
← few-shot examples
chosen by prompt optimization

```
Sentence: <new sentence>
Label: ______
```

← new input

Few-shot prompting: Brown et al. 2020, Language Models are Few-Shot Learners · Labeling guide: Shah et al. 2023, Table 10

Prompt optimization (GEPA)



GEPA’s prompt, before and after

Before · the paper’s prompt, 71 words

Discard all the previous instructions. Behave like you are an expert sentence classifier. Classify the following sentence from FOMC into **'HAWKISH'**, **'DOVISH'**, or **'NEUTRAL'** class. Label **'HAWKISH'** if it is corresponding to tightening of the monetary policy, **'DOVISH'** if it is corresponding to easing of the monetary policy, or **'NEUTRAL'** if the stance is neutral. Provide the label in the first line and provide a short explanation in the second line.

After GEPA · 576 words, excerpt

You are an expert sentence classifier for FOMC statements.

HAWKISH – the sentence signals tightening of monetary policy:
- Inflation is expected to rise or is rising.
- Unemployment is falling or is expected to fall.
- Economic growth is strong or projected to be high. …

DOVISH – the sentence signals easing of monetary policy:
- Inflation is expected to fall or is falling. …

NEUTRAL – the sentence is descriptive or mixed, with no clear tightening or easing signal: …

If the sentence contains both dovish and hawkish cues, default to NEUTRAL unless one side dominates.

+ 6 labeled examples

Fine-tuning (LoRA)

Freeze the pretrained weights

W doesn't change.

Train two small matrices

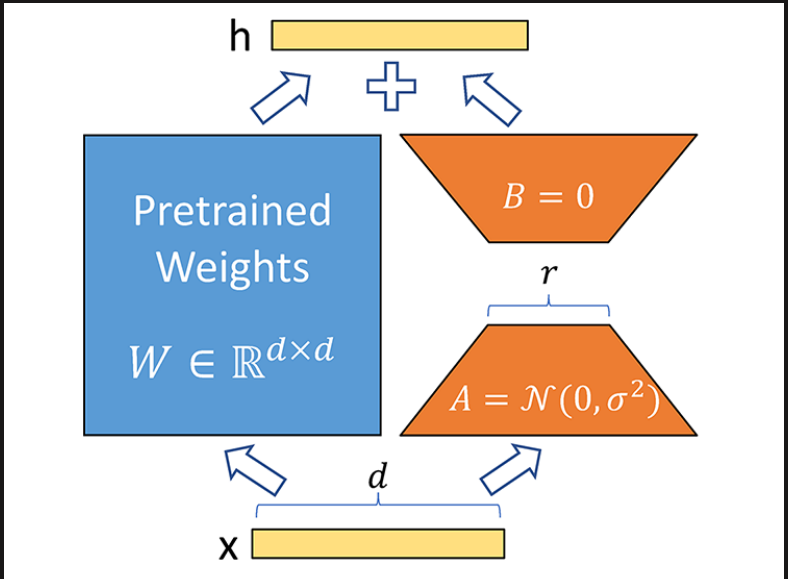
B × A is the update, with a small rank r.

Full update, d = 4096: 4096 × 4096 = 16.7M numbers

LoRA, r = 8: (4096 × 8) + (8 × 4096) = 65K, 256× fewer

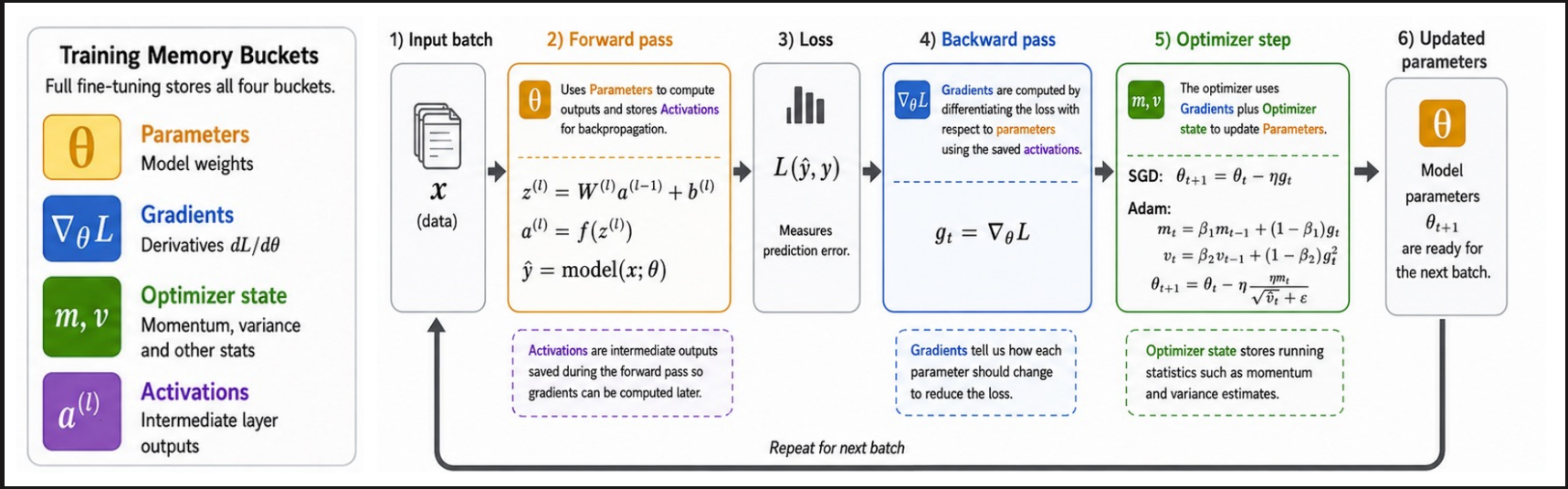
Learn from labeled examples

Same objective as full fine-tuning.



Hu et al. 2021, Figure 1

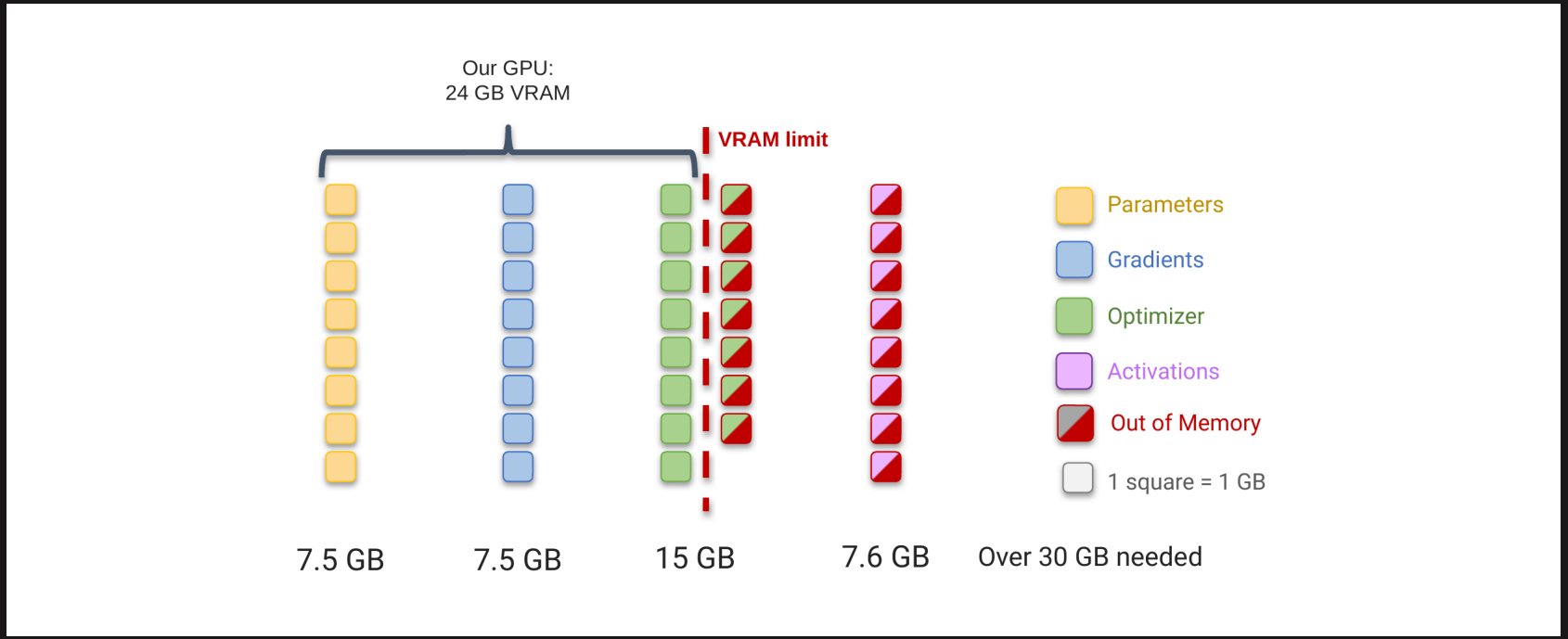
Where training memory goes



Off-the-shelf and GEPA train nothing: **parameters only, ~7.5 GB.**

Nemotron 3 Nano 4B in bf16: 4.04B weights × 2 bytes

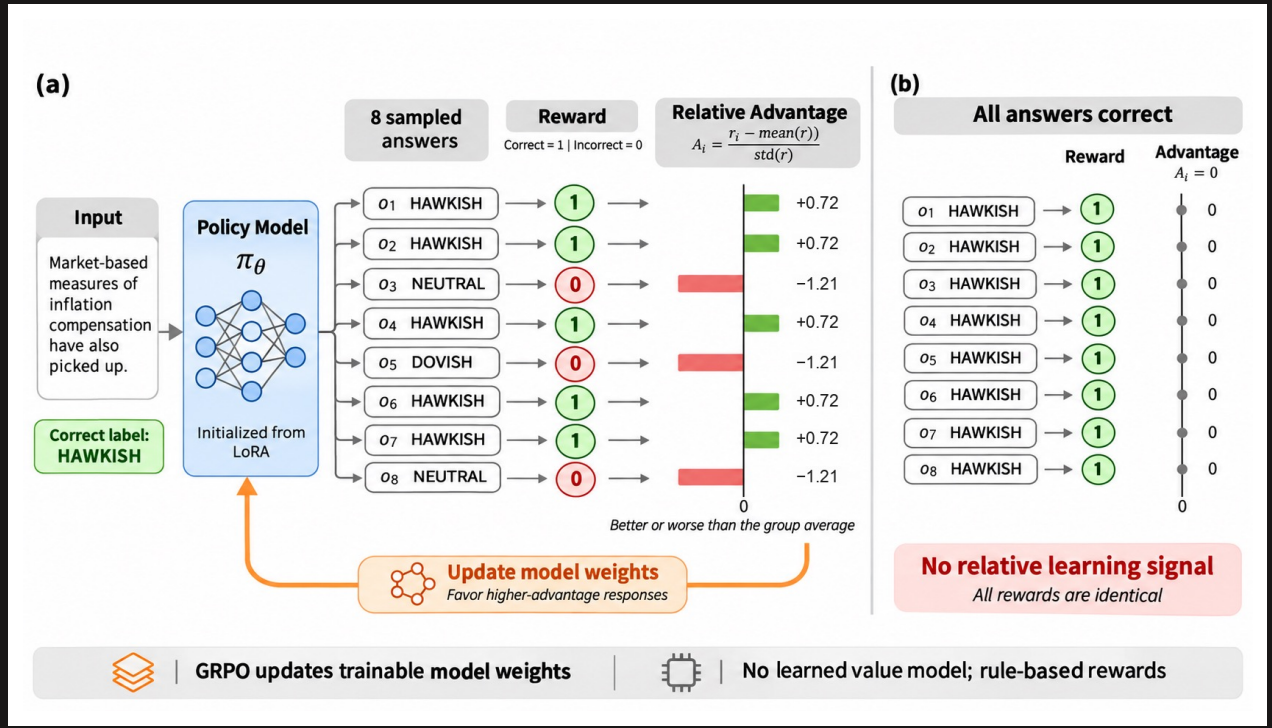
Why LoRA: memory efficiency on Nemotron 3 Nano 4B



Full fine-tuning: ~38 GB. LoRA: 9.0 GB. QLoRA: 5.4 GB.

bf16. LoRA and QLoRA measured; full fine-tuning estimated. · QLoRA in depth: PEFT talk (ODSC AI West 2026)

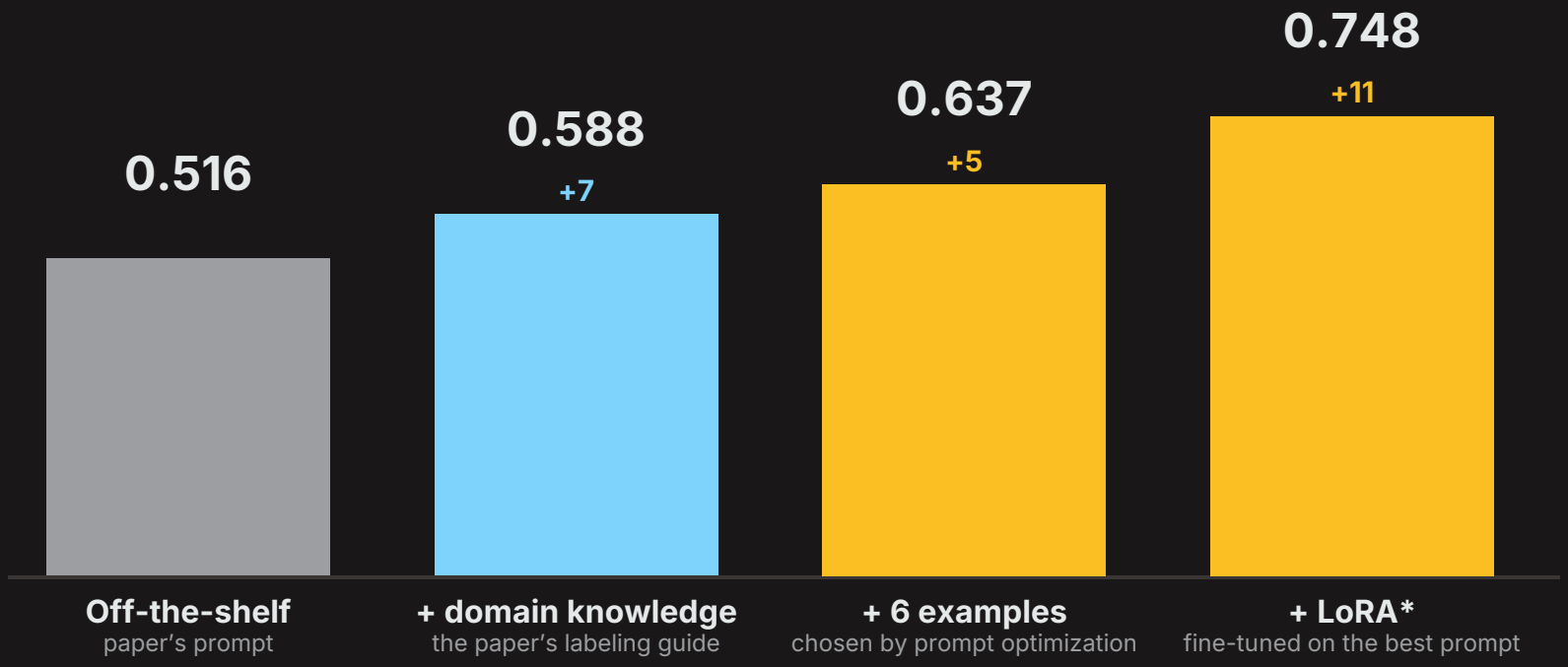
GRPO: Learning from Relative Rewards



GRPO compares multiple responses to the same input and updates model weights to favor better-performing answers.

Results: a better prompt, then fine-tuning

FED INTEREST RATES · Nemotron 3 Nano 4B · test weighted F1, mean of 3 seeds



*RL (GRPO) on top added little: after fine-tuning, most groups of 8 tries all score the same, which gives RL no signal.

Where prompt optimization shines

WHEN TO REACH FOR IT · reported results from teams using GEPA

You can't touch the weights	Multi-step agents	Rich text feedback	Few labels, costly rollouts
Models behind an API, e.g. on Amazon Bedrock. The prompt is the only knob.	Tunes every prompt in the pipeline together.	Compiler errors, test output, a judge's reasons.	Works from 20–100 examples. Far fewer rollouts than RL.
Nubank LLM judges: 69% → 89%	**Instacart assistant: 77% → 85%**	**GPU code compiles: 67% → 93%**	**vs GRPO: up to 35x fewer rollouts**

Sources: GEPA use-case showcase (gepa-ai.github.io); Decagon engineering blog, 2026; Agrawal et al., ICLR 2026.

From one laptop GPU to production on AWS

On one laptop GPU		On AWS
GEPA’s reviewer was the same 4B model	→	**Amazon Bedrock** **A larger model reviews the prompt**
Every seed and method ran one at a time	→	**Union on Amazon EKS** **Steps run side by side, each sized to fit: CPU, an NVIDIA GPU slice (MIG), or a full GPU**
24 GB: no full fine-tuning, no larger Nemotron	→	**Amazon EC2 GPU instances, Spot** **Bigger GPUs only for the steps that need them, scaled up and down automatically**
Laptop standby stopped a training run	→	**Union** **Interrupted runs recover without repeating completed work**

Try Union on AWS Marketplace

Union, in your own AWS account

Prompt optimization, fine-tuning and RL as one pipeline

Interrupted runs resume without repeating finished work

Self-serve on AWS Marketplace · 30-day free trial

Union on AWS Marketplace

Thank you. Questions?